Fourier Transform Based Spectrum Analyzer Simulation

This program provides a complete simulation of an end to end spectrum analyzer and includes the following functionality:

- Sampling rate 44.1 kHz
- Variable FFT frame length (default = 1024 samples)
- Generate a sine wave with variable amplitude (default = 1V) and frequency (default = 1 kHz)
- N (default = 8) bit Analog to Digital Converter (ADC) with a hard limit at Full Scale Deflection (FSD)
- Window (default = Hann)
- FFT
- Square magnitude
- One-pole filter with normalized gain of 1 (default alpha = 0.9)
- dB conversion with 0dBFS = Full Scale Deflection of the Analog to Digital Converter
- Scaling / plotting

The default input voltage (1V) of the sine wave maps to the FSD of the ADC and the FFT output is scaled so that FSD into the ADC equals 0 dBFS (dB Full Scale) in the graph.

I.E. for the default values: -1V maps to -128 and +1V maps to +127.

The core code for the simulation can be found in the function update_data_and_plots().

Dependencies

This application requires the following Python packages to be installed:

- numpy
- scipy
- sys
- PyQt6
- pyqtgraph

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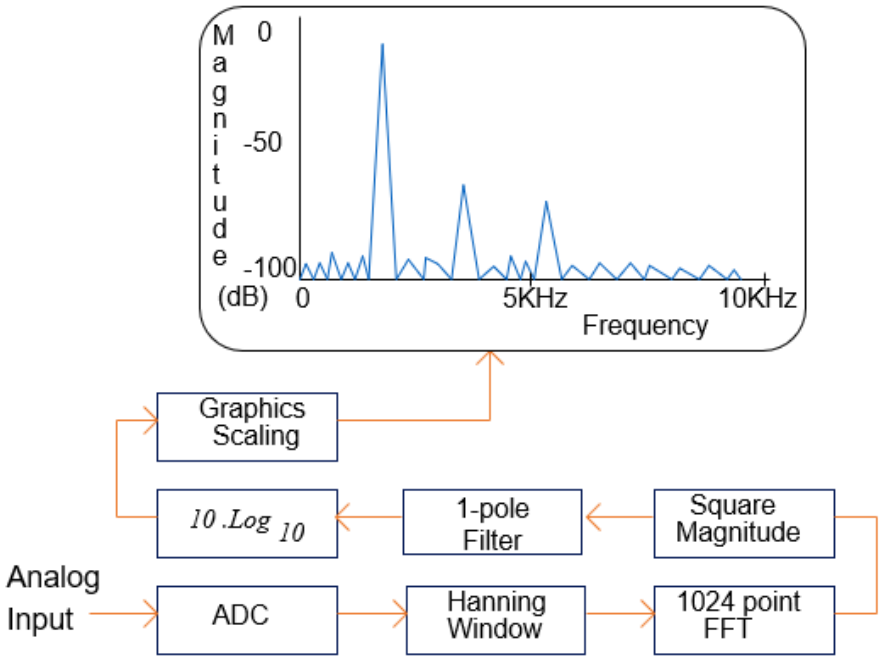


Figure 1: Spectrum Analyzer Simulation